Inside LLMs

Hands-On Classwork — How AI Predicts, Focuses, and Sometimes Gets Fooled

Name: 　　　　　　　　Date:

Section 1: Setup & Big Idea (~5 min)

LLMs are not magic brains. They are pattern-learning systems that build text one token at a time.

Which AI platform are you using today? (ChatGPT, Gemini, Claude, Grok, or another tool)

Before we start: In your own words, what do you think happens inside an AI when it answers a prompt?

Section 2: Tokenization — Text Becomes Pieces (~10 min)

Tokens are the small chunks of text an AI reads and writes. A token can be a word, part of a word, punctuation, or an emoji.

Activity 2A: Human Token Chopping

Draw slashes where you think an AI might split each phrase. There is not one perfect answer because different AIs use different tokenizers.

1. I love pizza!
2. unbelievable
3. ChatGPT is helpful.
4. The Minecraft creeper exploded.
5. antidisestablishmentarianism

Write your token guesses for the five phrases above:

Activity 2B: Compare With an AI

Ask an AI: 'Explain how you might tokenize these phrases for a beginner.' What did it say that surprised you?

Why can unusual names, slang, emojis, or mixed languages be harder for an AI to process?

Section 3: Next-Token Prediction — Human Autocomplete (~10 min)

An LLM repeatedly asks: 'Given everything so far, what token probably comes next?' Then it adds that token and keeps going.

Activity 3A: Predict First

Sentence starters:
A. For breakfast I ate...
B. The dragon opened the door and saw...
C. The best way to study for a test is...
D. In the year 2050, schools will...

For each starter, write what YOU predict the next few words will be:

Activity 3B: Check the AI

Give the same starters to an AI. Which answers matched your predictions? Which were different?

What does this show about 'likely' answers versus 'guaranteed' answers?

Section 4: Temperature — Safe vs. Wild Choices (~10 min)

Temperature is like a creativity dial. Low temperature usually gives safer, more predictable text. Higher temperature can be more creative but also more random.

Use the same prompt twice if your AI lets you choose creativity/temperature. If it does not, ask it to answer once 'very predictably' and once 'very creatively.'
Prompt: Write a 4-line story about a robot discovering a mysterious box at school.

Low creativity / predictable version:

High creativity / surprising version:

Compare them. Which version was clearer? Which was more fun? Which one would you trust more for facts?

Section 5: Attention — What Parts Matter Right Now? (~10 min)

Attention helps the model use earlier tokens that matter for the next word. It is like highlighting the important clues in a prompt.

Activity 5A: Highlight the Clues

Prompt: Maya put the trophy on the shelf after the soccer game. Later, her little brother moved it into the blue backpack. Where is the trophy now?

Underline/highlight the words that matter most. Then answer the question:

Ask an AI the same question. Did it focus on the same clues as you? Explain.

Activity 5B: Distractor Challenge

Add one extra distracting sentence to the prompt that should NOT change the answer. What sentence did you add?

Did the AI stay focused or get distracted? What does that teach you about writing prompts?

Section 6: Hallucination Detective (~10 min)

Because LLMs predict likely text, they can sometimes produce confident answers that are wrong. That is called a hallucination.

Try two tricky prompts:
1. What is the world record for most homework completed in one hour?
2. Give three facts about a famous scientist named Dr. Zorbius Mapleton.

What did the AI answer for prompt 1? Did it admit uncertainty or invent something?

What did the AI answer for prompt 2? What clues tell you whether this person is real?

How would you verify an answer before using it in homework, a video, or a post online?

Section 7: Reflection & Big Ideas (~5 min)

Connect what you learned today to prompt engineering, agents, and responsible AI use.

Explain LLMs in one sentence using the words token, predict, and context.

Which activity helped you understand AI the most? Why?

What is one thing you will do differently when prompting AI after today?

Rate today's classwork from 1 to 5 and explain your rating in one sentence: